technische universität
dortmund

Poverty as a Cardiometabolic Risk Factor:
Direct Effects from Dietary Quality and Physical Activity Engagement in U.S. Adults (NHANES 2007–2018)

Data and AI in Economics

Group R
Harsh Der, Avi Gupta, Priyanshu Kaushal

technische universität
dortmund

Motivation and Economic Relevance

- Type 2 Diabetes and Cardiometabolic diseases (T2D, CVD) drive immense healthcare costs and early mortality.
- While disease risk naturally increases with age, does poverty create a distinct socioeconomic gradient that significantly accelerates or exacerbates this risk?
- Does the Family Poverty Income Ratio (PIR) has a direct effect on Type 2 Diabetes (T2D) and cardiovascular disease (CVD) among U.S. adults aged 20 years and older?
- Or whether the relationship is mainly explained through lifestyle-related pathways such as diet quality and physical activity levels?

technische universität
dortmund

Data & Key Variables

- National Health and Nutrition Examination Survey (NHANES) Data 2007–2018 (6 pooled cycles).
- **Analytic Sample:** *N = 27,713* complete cases of U.S. adults (aged ≥ 20).
- **Exposure Variable:** Family Poverty Income Ratio (PIR), segmented into quartiles from Q1 (Poorest) to Q4 (Richest).
- **Target Outcomes:**
  - **Type 2 Diabetes (T2D):** 16.1% overall prevalence.
  - **Cardiovascular Disease (CVD):** 8.7% overall prevalence.
- **Behavioral Mediators:** Composite Dietary Risk Score (based on sodium, saturated fat, carbs, fiber) and Physical Activity indicator.
- **Key Confounders:** Age, BMI, Gender, Race/Ethnicity, and Education.

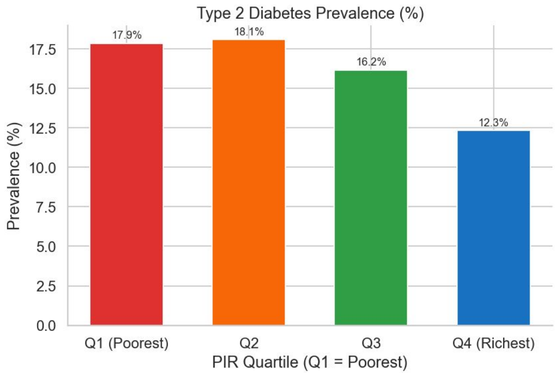

Figure 1: Type 2 Diabetes Prevalence across wealth brackets

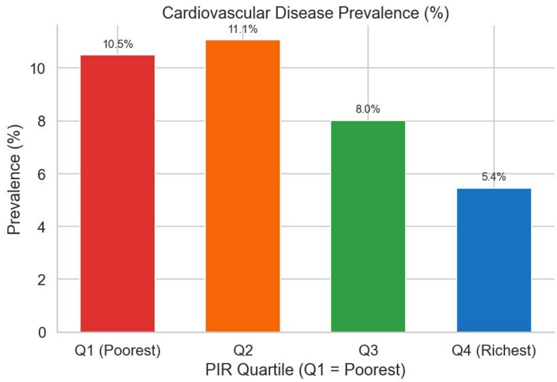

Figure 2: Cardiovascular Disease Prevalence across wealth brackets

Limitations

- **Cross-sectional design**: temporal precedence cannot be confirmed. Longitudinal data (CARDIA, Add Health) needed to establish causal ordering.
- **Reverse causality**: illness may cause poverty via medical costs and reduced labour capacity, upward-biasing the ACE.
- **Unmeasured confounders**: neighbourhood deprivation, chronic stress, food desert access absent from NHANES. E-values bound required confounder strength at RR >= 1.35 (T2D) and >= 1.75 (CVD).
- **Dietary measurement**: single 24-hour recall has high within-person variability, likely attenuates indirect effect estimates.
- **Class imbalance**: T2D (16.1%) and CVD (8.7%) as minority classes reduce supervised model recall. Future work: SMOTE or weighted loss.
- **Linear probability model**: OLS on binary outcomes can predict probabilities outside [0,1]. Logistic regression as robustness check.

technische universität
dortmund

Results

Causal Inference

- **PIR's Effect on T2D**: A 1-unit increase in PIR reduces the probability of Type 2 Diabetes by 1.09 percentage points ($p < 0.001$).
- **PIR's Effect on CVD**: A 1-unit increase in PIR reduces the probability of Cardiovascular Disease by 1.60 percentage points ($p < 0.001$).
- **Behavioral Mediation**: Diet quality and physical activity explain very little of these effects (< 2%), suggesting that the relationship between income and health outcomes is largely direct.

Supervised Learning

- Tuned Random Forest achieved an AUC-ROC of 0.758 for T2D prediction and a minority class recall of 0.74.
- Top Feature Importances for T2D:
  - Age: 0.567
  - Dietary Risk: 0.145
  - Poverty Income Ratio : 0.126
  - Physical Activity: 0.031

Unsupervised Learning

- **Socioeconomic Penalty Low-income** individuals with high BMI show a T2D prevalence of 28.5%, which is 6 times higher than younger, healthier low-income individuals at 4.6% T2D prevalence.
- **Affluence as a Buffer:** Despite having lower physical activity levels (39.3% active) than the 'High BMI' group (43.6% active), the 'Affluent' cluster maintains significantly lower T2D prevalence (11.9%) compared to the 'High BMI' cluster (28.5%), demonstrating wealth's protective effect.

technische universität dortmund

Conclusion

Causal Inference	Supervised Learning	Unsupervised Learning
Via backdoor adjustment, robustly establishes that higher Family Poverty Income Ratio (PIR) has a significant direct protective causal effect on both Type 2 Diabetes and Cardiovascular Disease; crucially, this effect is minimally mediated by dietary quality or physical activity	Dietary Risk and the Poverty Income Ratio have strong predictive association with cardiometabolic risk; however, this predictive capability, while highlighting key risk factors, does not establish causal direction or disentangle direct effects from mediation.	While age is a primary driver of cardiometabolic risk, particularly for the senior citizens, both high BMI in lower-income groups and the 'structural buffer' of affluence independently and significantly shape disease prevalence, with wealth offering protection even amidst less active lifestyles.

Hence we can conclude that, Family Poverty Income Ratio (PIR) exerts a statistically significant direct causal effect on cardiometabolic disease risk (T2D and CVD) among U.S. adults, largely independent of behavioral pathways like diet quality and physical activity; while supervised models identify age, dietary risk, and PIR as key predictors, the causal analysis reveals minimal mediation by these lifestyle factors, and unsupervised clustering further elucidates the heterogeneous impact of poverty, highlighting distinct high-risk phenotypes thereby underscoring the predominant role of **structural socioeconomic factors over individual behavioral choices in shaping health disparities**.

technische universität dortmund

References

- Marmot, M. (2005). Social determinants of health inequalities. The Lancet, 365(9464), 1099–1104.
- Stringhini, S. et al. (2017). Socioeconomic status and the 25x25 risk factors as determinants of premature mortality. The Lancet, 389(10075), 1229–1237.
- Darmon, N. & Drewnowski, A. (2008). Does social class predict diet quality? American Journal of Clinical Nutrition, 87(5), 1107–1117.
- VanderWeele, T.J. & Ding, P. (2017). Sensitivity analysis in observational research: introducing the E-value. Annals of Internal Medicine, 167(4), 268–274.
- CDC / NCHS. National Health and Nutrition Examination Survey (NHANES) 2007–2018. U.S. Department of Health and Human Services. Available at: https://www.cdc.gov/nchs/nhanes/
- American Diabetes Association (2022). Economic costs of diabetes in the U.S. in 2022. Diabetes Care, 46(7), 1230–1245.
- Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32.
- Baron, R.M. & Kenny, D.A. (1986). The moderator–mediator variable distinction in social psychological research. Journal of Personality and Social Psychology, 51(6), 1173–1182.
- Hill, A.B. (1965). The environment and disease: association or causation? Proceedings of the Royal Society of Medicine, 58(5), 295–300.

technische universität
dortmund

Appendix: Causal Inference

Effect	beta (ACE)	95% CI	p-value
Total: PIR->T2D	-0.0109	[-0.0137, -0.0081]	< 0.001
Direct: PIR->T2D	-0.0111	[-0.0139, -0.0083]	< 0.001
Indirect (T2D)	+0.0002	—	—
Total: PIR->CVD	-0.0160	[-0.0182, -0.0139]	< 0.001
Direct: PIR->CVD	-0.0161	[-0.0182, -0.0140]	< 0.001
Indirect (CVD)	+0.0001	—	—

Table 1: Average Causal effect

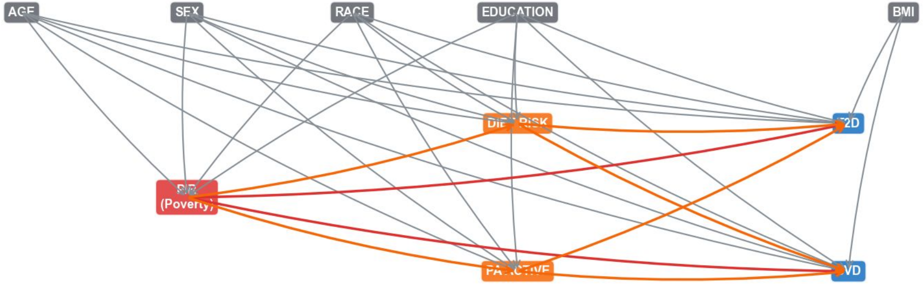

Figure 3: Causal DAG

Key Interpretations:

- Direct effect dominates: each 1-unit PIR rise reduces T2D by 1.09 pp, CVD by 1.60 pp
- Mediation is negligible: diet + PA explain -1.6% (T2D) and -0.4% (CVD) of the effect
- Suppression effect: controlling for mediators slightly amplifies the direct poverty effect
- E-values of 1.35 (T2D) and 1.75 (CVD) indicate robustness to unmeasured confounding

technische universität dortmund

Appendix: Supervised Learning

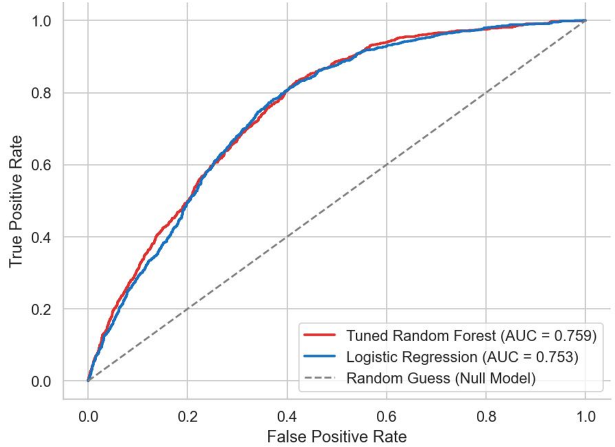

Figure 4: ROC Curve Comparison

	Precision	Recall	F1- score	Support
0	0.84	0.99	0.91	4651
1	0.47	0.05	0.09	892
Accuracy			0.84	5543

Table 2: Classification report Random Forest

1. Hyperparameter Optimization

GridSearchCV optimized across 5 cross-validation folds for AUC-ROC scoring:

{'max_depth': 10, 'min_samples_split': 5, 'n_estimators': 200}

2. Feature Importance

- **Age:** 56.7% — Dominant predictor of cardiometabolic risk
- **Dietary Risk Score:** 14.5% | **Poverty Income Ratio:** 12.6%
- **Physical Activity:** 3.1% | **Education Level:** 3.0%

Appendix: Unsupervised Learning

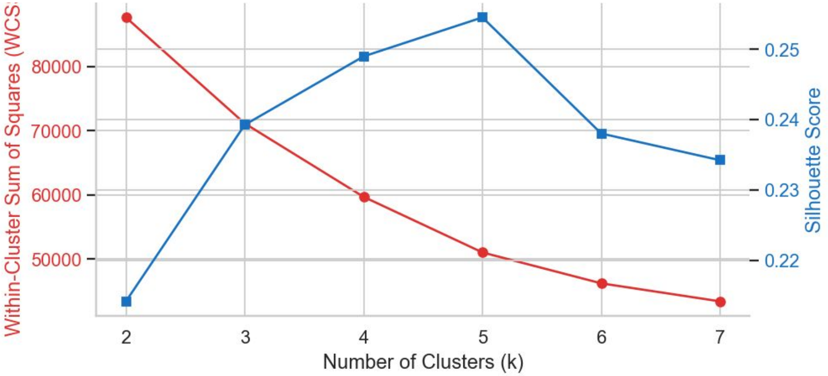

Figure 5: K means Tuning

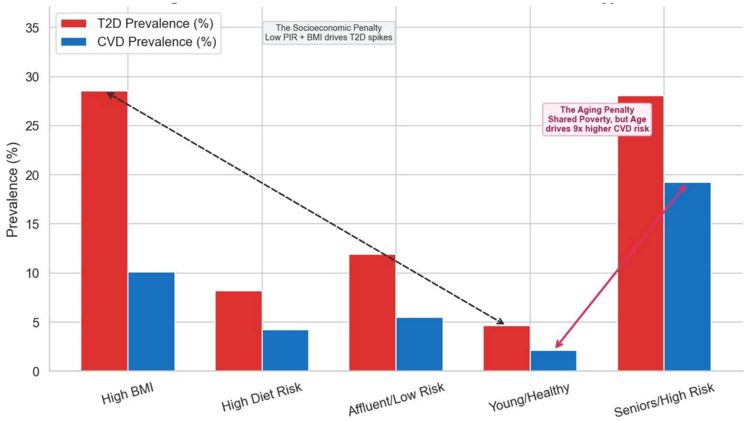

Figure 6: Disease Prevalence across Refined Health Phenotypes

The fact that the Affluent Cluster (2) has significantly lower disease rates than the High BMI Cluster (0), despite the Affluent group being more sedentary, suggests that wealth provides a "structural buffer" against metabolic disease that lifestyle choices do not fully offset.

Cluster	Count	Wealth (PIR)	Diet Risk	Age	BMI	% Active	T2D %	CVD %
0	3,466	1.98	0.00	45.99	42.38	43.6%	28.5%	10.1%
1	3,074	2.14	1.09	39.51	27.95	56.7%	8.2%	4.2%
2	7,480	4.62	-0.11	51.06	27.65	39.3%	11.9%	5.5%
3	6,454	1.43	-0.12	33.27	26.33	47.1%	4.6%	2.1%
4	7,239	1.64	-0.24	67.96	28.02	32.6%	28.0%	19.2%